

Shashank Pratap Singh

Department of Mathematics, University of Iowa
[shashankpratapsingh.github.io](https://github.com/shashankpratapsingh) | [✉ shashank-singh@uiowa.edu](mailto:shashank-singh@uiowa.edu)

Profile

Ph.D. candidate in Mathematics (Expected 2026) specializing in representation theory, Hopf algebras, and tensor categories, with substantial experience as instructor of record in coordinated multi-section undergraduate courses. Research focuses on generalizing classical results such as Gabriel's Theorem to tensor categories.

Research Interests

- **Primary Areas:** Representation theory, Hopf algebras, Tensor categories
- **Additional Interests:** Quiver representations, Category theory
- **Specific Focus:** Generalizations of Gabriel's theorem, Taft algebras, module categories, algebraic structures in tensor categories

Education

Ph.D. in Mathematics, University of Iowa Expected 2026
Advisor: Prof. Ryan Kinser
Dissertation: "Quivers and Tensor Categories: Toward a Generalization of Gabriel's Theorem"

B.S & M.S. in Mathematics,
Indian Institute of Science Education and Research, Pune, India 2018
Master's Thesis: "Class Field Theory and Tate's Thesis"

Publications

- **In preparation:** "A Quiver-Theoretic Description of Taft Equivariant Morita Contexts"
- **Dissertation:** "Quivers and Tensor Categories: Toward a Generalization of Gabriel's Theorem"
(Draft available upon request)

Presentations

Conference Presentations

- **Cyclic quivers, Taft algebras, and the classification of bimodules**
Maurice Auslander Centenary Lectures and International Conference 2026 April 2026
- **A Quiver-Theoretic Description of Module Categories over Taft Algebras**
Seattle Noncommutative Algebra Conference 2025, University of Washington December 2025

- **Module categories in the representation category of Taft algebras**
QuaSy-Con III, Iowa State University November 2025
- **Algebra objects in the representation category of Taft algebras**
AMS Sectional Meeting, St. Louis, MO October 2025
- **Tensor Algebras in the Representation Category of Taft Algebras**
Hopf25, ULB Brussels April 2025
- **Understanding Hopf Algebra Representations through Quivers**
AMCS-MATH Day, University of Iowa September 2024

Seminar Talks

- **Tensor Algebras in Tensor Categories**
Algebra Seminar, Saint Louis University November 2024
- **Tensor Algebras in Tensor Categories**
Algebra Seminar, University of Iowa October 2024
- **Exact Module Categories and Module Algebras**
Algebra Seminar, University of Iowa February 2024

Outreach and General Audience Talks

- **Scissors, Shapes, and Tensor Products**
GAUSS, University of Iowa May 2025
- **The ABCs of Algebraic Geometry**
GAUSS, University of Iowa October 2023

Pedagogical Training

- **Certificate in College Teaching**, University of Iowa December 2025
Completed coursework in curriculum design, assessment strategies, and inclusive teaching practices

Teaching Experience

Instructor of Record, University of Iowa

- **College Algebra** (Spring 2023, Fall 2023, Fall 2024, Fall 2025)
Taught in a coordinated multi-section course (typically 15–20 sections) with 35–40 students per section. Collaborated with other instructors to maintain consistent standards and assessments across all sections
- **Elementary Functions** (Fall 2022)
Taught full course with 35–40 students, responsible for all aspects of instruction

Teaching Assistant, University of Iowa

- **Introduction to Abstract Algebra** (Spring 2025)
Led discussion sections and provided individualized student support
- **Engineering Math I: Single Variable Calculus** (Spring 2022)
Conducted discussion sessions and graded assignments
- **Qualifying Exam Preparation Seminars (Algebra)** (Summer 2023)
Organized and led intensive review sessions for graduate students

Tutoring and Academic Support

- **Math Platoon** (Spring 2022, Spring 2025, Fall 2025)
Provided specialized tutoring support for army veterans enrolled in university mathematics courses
- **Math Tutorial Lab** (Spring 2022, 2023, 2025; Fall 2022, 2023, 2024, 2025)
Delivered comprehensive tutoring services to students across various mathematics courses

Technical Skills

- **Mathematical Software:** SageMath, $\text{\LaTeX}$
- **Programming:** Python, C++
- **Mathematical Typesetting:** Advanced $\text{\LaTeX}$, TikZ, Beamer

Awards and Fellowships

- **Erwin and Peggy Kleinfeld Scholar,**
Department of Mathematics, University of Iowa
Fall 2023–Spring 2026
- **Kleinfeld Math Graduate Recruitment Fellowship,**
Department of Mathematics, University of Iowa
Fall 2021–Summer 2022
- **INSPIRE Scholarship,**
Department of Science & Technology, India
Fall 2013–Spring 2018

Service and Outreach

- **Directed Reading Program** (Fall 2024)
Mentored undergraduate student Grant Trentz in Cryptography and Number Theory
- **Directed Reading Program** (Spring 2026)
Mentored undergraduate student Hwanhee Byun in Category Theory & Representation Theory
- **Graduate Student Reading Group Organizer**
Selected materials and coordinated weekly discussions for 6–8 graduate students:
 - Hopf Algebras (Spring 2024)
 - Lie Algebras and their Representations (Fall 2023)
 - Tensor Categories (Spring 2023)